Status	Finished
Started	Tuesday, 20 October 2026, 12:23 PM
Completed	Tuesday, 20 October 2026, 12:32 PM
Duration	8 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Arranging data in a specific order
c. Removing duplicates from a list
d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Sorting data sequentially
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It makes the search operation slower
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sort()
- c. order()
- d. sorted()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Merge Sort
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Selection sort
- b. Quick sort
- c. Merge sort
- d. Heap sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Quick Sort
- c. Insertion Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **13**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. sorted()
c. sort()
d. arrange()

The correct answer is: sort()

Question **14**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Arranging data in a particular format
b. Inserting data into a list
c. Searching for data in a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It makes the analysis slower
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates